Testing quantum memories on IBM hardware

Nadav Abramson

1. THE EXPERIMENT

Two questions: how well do IBM's calibration data predict the errors of a logical quantum memory, and does a heavy-hex code get better when its distance goes from 3 to 5?

On `ibm_fez` the experiment ran fifteen jobs over five days: two placement pairs, X and Z memory, and 2500 shots per memory circuit at 1, 2, 3, 4, 6 and 8 rounds. Before each job, a prediction built from that day's calibration was computed and saved. None of its noise rates came from the job's own memory data, and no shot was discarded.

The short answer: the model underestimates the failures in both memories, and distance 5 does worse than distance 3 on these placements, as the model expected. The rest of the project looks for the missing error, with decoder checks, a channel-by-channel error budget, persistent readout patterns and a follow-up run with a modified circuit on `ibm_kingston`. Every result is about these chip regions and these circuits, not about heavy-hex codes in general.

2. CODE AND CIRCUITS

On IBM's heavy-hex lattice each qubit has at most three neighbors. The heavy-hex subsystem code combines surface-code and Bacon–Shor checks [1, 2] and is a $[[d^2, 1, (d-1)^2/2, d]]$ code (data qubits, logical qubits, gauge qubits, distance). In the X/Z convention of Ref. [3], its gauge group is

$$\begin{aligned}\mathcal{G} = \big\langle\, & X_{i,j}X_{i,j+1}, \\ & Z_{i,j}Z_{i+1,j}Z_{i,j+1}Z_{i+1,j+1} \quad (i+j \text{ odd}), \\ & Z_{i,1}Z_{i+1,1}\ (i \text{ odd}), \quad Z_{i,d}Z_{i+1,d}\ (i \text{ even}) \,\big\rangle,\end{aligned} \tag{1}$$

where i, j label data rows and columns, and indices stop at $d-1$ wherever an index plus one appears. The stabilizers are generated by

$$\begin{aligned}\mathcal{S}_X :\ & X_{i,j}X_{i+1,j}X_{i,j+1}X_{i+1,j+1} \quad (i+j \text{ even}), \\ & X_{d,j}X_{d,j+1}\ (j \text{ odd}), \quad X_{1,j}X_{1,j+1}\ (j \text{ even}), \\ \mathcal{S}_Z :\ & \textstyle\prod_{j=1}^{d} Z_{i,j}Z_{i+1,j}, \quad i = 1, \ldots, d-1,\end{aligned} \tag{2}$$

and the logical operators are $X_L = \prod_i X_{i,1}$ and $Z_L = \prod_j Z_{1,j}$. Each X stabilizer is one X gauge or a product of two. Each Z stabilizer is the product of the Z gauges between two adjacent rows; it has weight $2d$, and this Bacon–Shor half has no threshold [1, 4].

In X memory all data qubits start in $|+\rangle$ and are read in X after n rounds of gauge measurements; a shot fails if the decoded X_L differs from its initial value. Z memory starts in $|0\rangle$ and reads in Z. Z errors fail X memory and X errors fail Z memory.

Measuring the gauges. Each round measures all gauges of the memory basis, then those of the other basis. An X gauge needs one ancilla and two CX gates. A bulk Z gauge acts on four data qubits that its ancilla cannot reach directly, so two neighboring X ancillas act as flags: they collect the data parities, pass them to the Z ancilla and are uncomputed (Fig. 2a). Edge Z gauges reach their far data qubit through a relay. Without faults, flags and relays read 0. The distance-3 patch uses $9+6+4+4 = 23$ physical qubits and distance 5 uses $25+20+12+8 = 65$. The twelve-CX bulk gadget has four more CX gates than the flag circuits of Refs. [1, 3]. Any single fault leaves at most one data error, or two in the same column, up to the measured gauge, so dropping the flag outcomes does not reduce the X-memory circuit distance.

Detectors and decoding. If $m_{s,t}$ is stabilizer s in round t, the XOR of its gauge outcomes, the detector

$$D_{s,t} = m_{s,t} \oplus m_{s,t-1} \tag{3}$$

fires when the stabilizer changes. Memory-basis stabilizers give one detector per round and one at the final data readout; the first compares with the value fixed by preparation. Z memory also uses the flag and relay readouts as detectors. Before the first round the other-basis gauges are measured once, so every round runs the same gate sequence; those outcomes enter no detector.

Stim builds the detector error model and PyMatching decodes it by minimum-weight perfect matching [5–7]. Device and simulated shots are decoded with the same priors, those of the stored prediction. The circuit distance, the fewest undetected faults that flip the logical value, is d in both bases; Sec. 5 looks at one catch in the distance-5 Z matching graph.

Timing and placement. The circuits use CZ, $\sqrt{X}$, X and virtual R_Z gates with explicit waits. Waits of 500 ns or more after a qubit's first gate get two X pulses, at one quarter and three quarters of the wait; IBM's own decoupling and twirling were off. A round takes about 7.9 µs. Readout and reset take 6.7 µs of it on the critical path, so the data qubits sit idle for 94–97% of the round (Fig. 2b).

Placements were ranked by an estimated error per round,

$$E = \sum_{\text{CZ}} r_{qq'} + \sum_{\text{meas., reset}} e_q + \sum_{\text{data}} \frac{9\,\mu\text{s}}{2T_{1,q}}, \tag{4}$$

with gate and readout errors $r_{qq'}$ and e_q, and 9 µs a pre-compilation estimate of the round time. Every day had

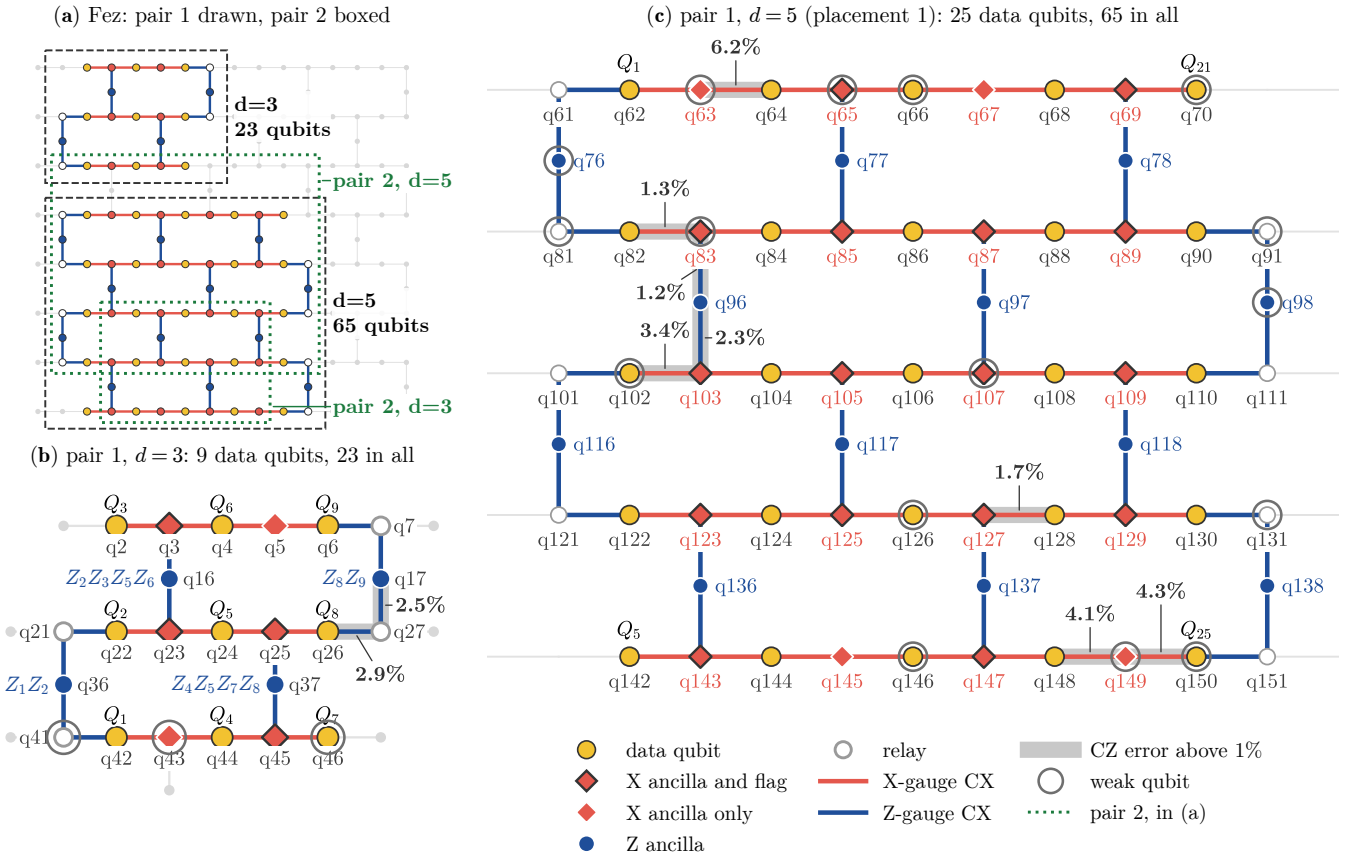


FIG. 1. Fez layouts. (a) Pair 1 (black boxes) and pair 2 (green boxes, separate jobs). (b,c) Pair 1's distance-3 and distance-5 patches; Q labels are data qubits and q labels are physical qubits. Shaded couplers have CZ error above 1%; rings mark readout error above 5%, $T_1 < 50\,\mu s$ or $T_2 < 30\,\mu s$. The large patch contains weak parts, including q149 and q150 with a reported T_2 of only 5–6 µs.

eighteen usable distance-3 placements but only two usable distance-5 placements, the same two each day, sharing 51 of 65 qubits. A 5% CZ-error cutoff or a 10% readout-error cutoff would have left no distance-5 placement at all. The selected large patches include CZ errors of 3.4–7.0% and readout errors of 8–14%, and one relay reached 48.6% on day 4.

Pair 1 ran both distances at once on disjoint patches, on the same qubits every day. Pair 2's small patch ran about a minute after its large one and overlapped it. So the two large layouts and the five days are not independent placements. One pair on one day is a *pair-day*: ten pair-days, two memories and two distances give forty error curves.

3. NOISE MODEL AND IDLE TESTS

The model follows the submitted circuits gate by gate and wait by wait. IBM's average gate infidelities become depolarizing channels,

$$p_1 = \tfrac{3}{2}\, r_q, \qquad p_2 = \tfrac{5}{4}\, r_{qq'}, \tag{5}$$

spread equally over the 3 or 15 nonidentity Paulis [9]. Readout and reset both use the reported readout error e_q. A wait t gets Pauli-twirled relaxation and dephasing [10],

$$\begin{aligned} p_x = p_y &= \tfrac{1}{4}\big(1 - e^{-t/T_1}\big), \\ p_z &= \tfrac{1}{2}\big(1 - e^{-t/T_2}\big) - \tfrac{1}{4}\big(1 - e^{-t/T_1}\big), \end{aligned} \tag{6}$$

with IBM's T_2 capped at $2T_1$. During decoupled waits

$$\frac{1}{T_2^{\mathrm{DD}}} = \frac{1}{2T_1} + f\,\Gamma_\phi, \qquad \Gamma_\phi = \frac{1}{T_2} - \frac{1}{2T_1}, \tag{7}$$

so $f = 0$ removes pure dephasing and $f = 1$ keeps IBM's echo value; shorter waits use the capped T_2. A qubit ideally in $|0\rangle$ gets no idle channel until its next gate, which ignores relaxation after a reset error.

The model has no leakage, coherent errors, ZZ crosstalk, drift within a job or state-preparation errors. Isolated benchmarks may also miss errors that appear only in parallel operation or in mid-circuit readout and reset, while the single-qubit gates around each CZ may count error that its benchmark already includes. The bias can therefore go either way.

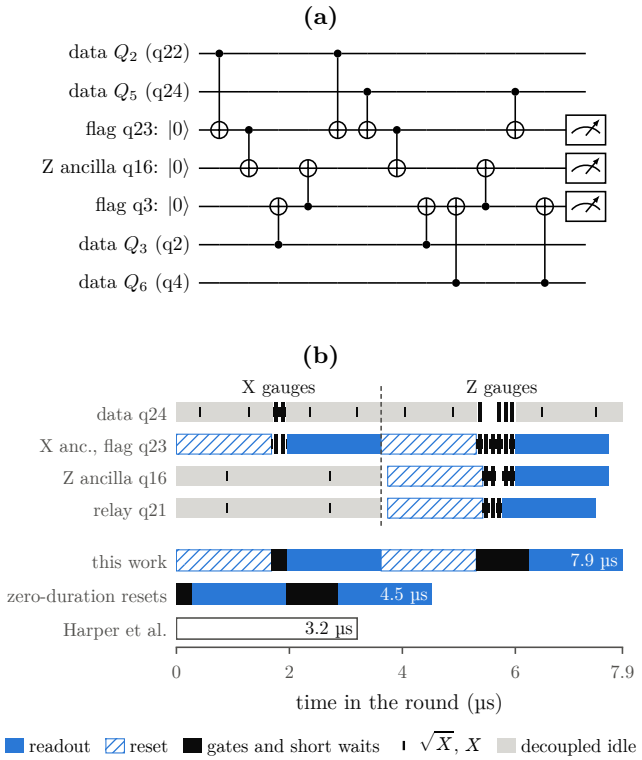


FIG. 2. (a) A bulk Z-gauge measurement through two flags and the Z ancilla, with twelve CX gates, each compiled as H·CZ·H. Without faults the flags return to 0. (b) One compiled Fez round, one qubit per role. Readout and reset dominate the 7.9 µs round; the bars below show the same round with zero-duration resets and the parallel, reset-free circuit of Ref. [8].

Each stored prediction used 200,000 simulated shots per circuit at $f = 1$; this *stored prediction* is the reference throughout. Each job also ran ten idle circuits, with the memory schedule but every CZ replaced by an equal wait, and read X and Y after 1, 2, 4, 8 and 16 rounds. The contrast

$$v(n) = \sqrt{\langle X\rangle^2 + \langle Y\rangle^2} \tag{8}$$

does not see frequency offsets, and each data qubit's f comes from

$$\ln v(n) = \alpha + \ln \bar{X}_0(n) - f\big[\ln \bar{X}_0(n) - \ln \bar{X}_1(n)\big], \tag{9}$$

where $\bar{X}_0$ and $\bar{X}_1$ are the exact model curves at $f = 0$ and $f = 1$ and α absorbs a contrast offset. The median f over data qubits gives a second, *in-job* prediction. This test sees the Z and Y errors that fail X memory, not the X flips that fail Z memory, and its ancillas stay near $|0\rangle$, unlike in a real memory.

Qubits do not share one f. A common f fails the per-qubit fit in every job (χ^2/ν = 12–80), and f grows with the reported T_2, with rank correlation 0.66 over 36 qubits (Fig. 3b). For q150, with a reported T_2 of 5–6 µs, the job's median f predicts a Z/Y error of 32–39% per

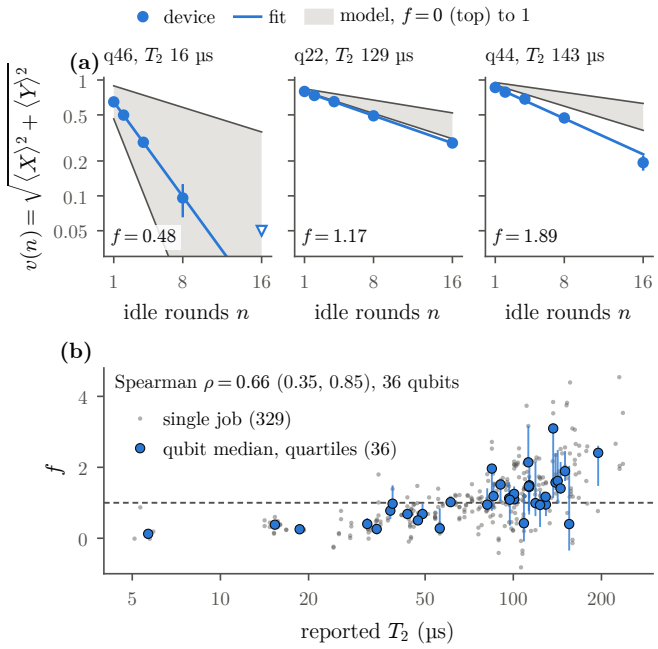


FIG. 3. Idle decay. (a) Three data qubits of day 1's distance-3 pair-1 patch, with the $f = 0$ and $f = 1$ model curves and the fit; the open point is below the 0.05 contrast cutoff. (b) Fitted f against IBM's reported T_2, with per-qubit medians and quartiles.

round, while that qubit's own fit gives 4–13% on four days. Per-qubit values of f change the comparison between placements but do not close the device–model gap. Writing the measured extra dephasing as $\Gamma_{\rm dd} = f\,\Gamma_\phi$, the qubit medians follow

$$\Gamma_{\rm dd} = 0.11\,\Gamma_\phi + 0.0071\,\mu{\rm s}^{-1}, \tag{10}$$

a floor of about 2.5% phase-flip probability over 7.1 µs of decoupled idle, or 1.6–1.8% with other qubit selections or weights. Both sides of this fit use the same calibration, so it describes the data rather than separating a physical channel.

Calibration checks. The calibration snapshots were 21–88 minutes old at the first job, but many values had not changed from earlier snapshots; on day 1 every value matched a snapshot 268 minutes old. The reported readout errors come in steps consistent with 4096 shots per prepared state, which means about 11% relative shot error at the median readout error. A recent snapshot is therefore not a fresh or precise measurement of every input.

Using IBM's smaller initialization error for the resets lowers the predicted Z-memory decay by 0.6–0.8 percentage points and mostly worsens the detector agreement; the larger reset and readout errors fitted in Ref. [8] overshoot. No single mid-circuit reset error fixes the discrepancy.

4. FEZ RESULTS

At one to four rounds the device fails more often than the stored prediction in all 160 combinations of day, pair, basis, distance and depth (Fig. 4). Averaging the pairs within each day and then the five days, the shortfall is 3.8–7.6 percentage points in X memory and 8.8–16.6 in Z memory. The 160 cells share qubits and days, so the intervals in Fig. 4 come from the five daily means; a few lose their zero exclusion when a day is left out, but every daily mean is positive.

To compare depths, each curve is fitted to

$$1 - 2P_{\mathrm{L}}(n) = A\,(1 - 2\varepsilon)^n \tag{11}$$

by binomial maximum likelihood. Here ε is an effective decay per round over the fitted depths and $\Lambda = \varepsilon_3/\varepsilon_5$; $\Lambda > 1$ would mean distance 5 helps. A is free, with $A(1-2\varepsilon) \le 1$. Twelve Z-memory fits have $A > 1$, so A is not a preparation contrast, and saturation near $P_{\mathrm{L}} = 1/2$ leaves little rate information at late depths.

Measured ε exceeds the stored prediction on 39 of 40 curves. The geometric-mean device/model factors are 1.27 and 1.19 in X memory at distances 3 and 5, and 2.06 and 1.84 in Z memory. Other fit windows give 1.15–1.28 in X and 1.65–2.10 in Z, and updated idle inputs do not remove the gap.

Distance and day. Distance 5 fails more often in X memory in 59 of 60 day–pair–depth combinations. With in-job priors, $\Lambda_{\mathrm{X}} = 0.67$ and 0.49 for the two pairs, both below one as predicted, although this comparison mixes distance with placement quality.

For pair 1 the predicted Λ_{X} swings between 0.54 and 0.82 over the five days, while the measured one hardly moves (Fig. 5a). The big swing is day 2, when four neighboring distance-3 qubits were reported as much worse than usual; setting them to their medians on the other days lowers the predicted Λ_{X} by 0.15. The device did not follow the reported change.

5. ERROR BUDGET AND DECODER CHECKS

Separating the model's channels. The model splits into CZ gates, single-qubit gates, readout and reset, idle relaxation, and idle dephasing; the idle channel factors into relaxation followed by a phase flip, so the last two can be varied separately. All 31 nonempty combinations were simulated for all fifteen jobs at 20,000 shots per circuit, decoded with the full model's decoder. The channels interact, so switching one off does not remove a fixed share of the logical error. The Shapley value [11] averages each channel's marginal effect over every order of introducing the five channels,

$$\phi_c = \frac{1}{5!}\sum_{\pi}\left[\varepsilon(P_\pi^c \cup \{c\}) - \varepsilon(P_\pi^c)\right], \tag{12}$$

where P_π^c holds the channels before c in the order π; the shares add up to the full model's ε (Table 1).

Idle noise accounts for about half of the modeled decay and the CZ gates for about a quarter. Raising the dephasing by about 1.5 fixes the X-memory summaries but cannot fix Z memory. A two-channel fit matches all four decay summaries with relaxation scaled by 2.63 and dephasing by 0.60, yet the detector rates still disagree, so the summaries do not identify the missing channel.

Scaling every error rate of the model by one factor k changes ε faster than linearly, with $d\ln\varepsilon/d\ln k = 1.3$–$1.4$ near $k = 1$, while Λ hardly moves (0.625 to 0.622 for pair 1). A ratio can stay put while the errors behind it change a lot.

Circuit distance and the matching graph. Fault searches confirm circuit distance 3 and 5. The distance-5 Z matching graph still has a logical path of only three edges; two of them appear because Stim decomposes data X errors that also fire flags and relays. To see whether this matters, the flag and relay detectors were dropped while every gate, readout and the logical observable stayed. The reduced Z graph has distance 3 or 5, and with at most twenty stabilizer detectors at depths 1–4, exact maximum-likelihood (ML) decoding is possible by summing the model probability of every syndrome and logical value. On the device shots, reduced matching changes the mean failure by -0.04 to $+0.24$ percentage points and exact ML by -0.92 to -0.34. The gap to the model stays at 8.5–16.8 points, positive in every pair-day.

For distance-3 X memory at depths 1–4, exact ML changes the device failure by -1.64 to $+1.16$ points and leaves gaps of 2.76–11.06 points. At four rounds it makes the same call as matching on 80% of the single-fault probability that matching "miscorrects", so most of those calls are what the assumed model prefers anyway.

Priors. Calibrated priors beat uniform weights. Learning the pair-edge priors from detector correlations [12] changes the X-memory decay only a little, and learning every edge type makes it worse; both are in-sample checks. None of the decoders tried closes the gap.

6. PERSISTENT FLAG AND RELAY READOUTS

Without faults a flag or relay reads 0. On Fez in Z memory, a 1 is followed by another 1 in the next round 23.0% of the time, against 8.2% after a 0; the model gives

TABLE 1. Shapley shares of the model's ε, in percent, averaged over the ten pair-days. They split the model's error, not the device's excess.

	CZ	1-qubit	Readout/reset	T_1 idle	Dephasing
X, $d = 3$	22	10	12	22	35
X, $d = 5$	25	7	17	17	34
Z, $d = 3$	25	8	13	53	0
Z, $d = 5$	30	9	16	46	0

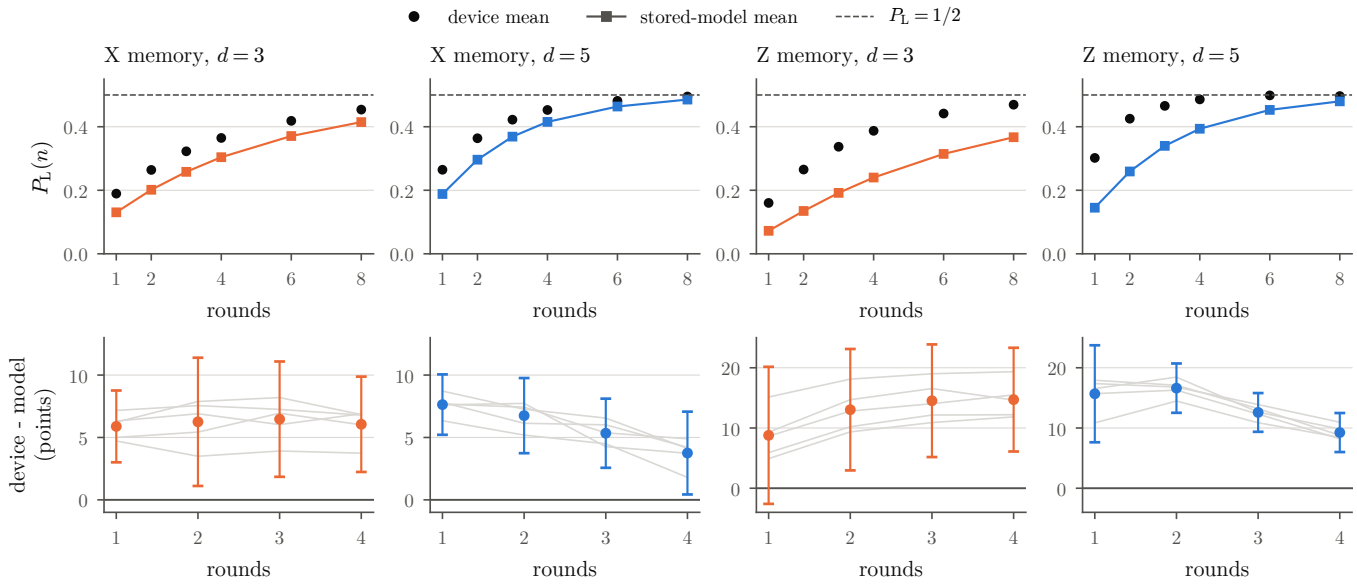


FIG. 4. Fez, decoded with the stored $f = 1$ priors on device and model. Top: failure probability averaged over days and placements; model points are joined, not fitted. Bottom: device minus model at one to four rounds. Gray lines are daily means; bars are approximate simultaneous 95% intervals from the five daily means (t with four degrees of freedom, Bonferroni over sixteen cells).

6.9% in both cases. For a binary readout x_t define the persistence

$$\Delta(\ell) = \Pr(x_{t+\ell} = 1 \mid x_t = 1) - \Pr(x_{t+\ell} = 1 \mid x_t = 0). \tag{13}$$

Over 320 series per basis the next-round persistence averages 14.8 percentage points in Z and 15.2 in X, against about zero in the model, and it decays with the lag (Fig. 5b). Rates that only differ from shot to shot would give a persistence independent of the lag, so they cannot explain this alone.

A hidden state. A two-state hidden Markov model [13, 14] enters a persistent state with probability a per round and leaves it with probability b; a readout is 1 with probability u_0 outside the state and u_1 inside. The four-, six- and eight-round Z-memory records give medians $a = 2.5\%$, $b = 22.3\%$ and $u_1 = 0.51$. With round time τ the state lives

$$T_{\mathrm{p}} = -\frac{\tau}{\ln(1-b)} \simeq 31\,\mu\mathrm{s}, \tag{14}$$

and its mean residence time τ/b is about 35 µs. Its occupation starts near 3%, grows toward 10% and averages 6.3% over these circuits; X memory gives similar rates. The state describes the readouts; it is not a measured transmon level.

Leakage could produce such persistence and spread it through CZ partners. A simple leakage simulation fitted to the detector statistics, however, did not reproduce the logical errors of both memories: depending on how a leaked qubit acts on its partner, it overshoots X memory or leaves a Z deficit.

Streaks and failures. After two consecutive flag or relay 1s, nearby X-memory stabilizers fire 8.4 percentage points more often two rounds later, but they already fire 5.6 points more often two rounds before the streak. Logical failure is 4.8 points higher in X memory, against 0.8 in the model, and at a fixed total of flag and relay 1s the streak association drops to 0.2 points. Streaks go together with failures without clearly causing them.

7. FOLLOW-UP ON KINGSTON

The follow-up ran on another Heron r2 processor, `ibm_kingston`, with three circuit changes: no mid-circuit resets, XY4 decoupling instead of paired X pulses, and randomized (twirled) CZ layers. Without a reset, an ancilla starts the next gadget in $X^m|0\rangle$, where m is its last outcome, so each gauge value is the XOR of the current outcome and known earlier ones. A round takes 5.8 µs at distance 3 and 6.0 µs at distance 5. Two memory jobs covered rounds 1–4, with 8192 shots per circuit in total and sixteen randomizations per job.

This time the placement score included T_2, but no distance-5 patch passed every cutoff, and the fallback still had CZ errors of 6.4% and 7.1%. The distance-3 patch sat inside the distance-5 footprint, with different roles and schedules. Chip, placement and circuit all changed at once, so the improvement below cannot be credited to any single change.

The fitted decay parameters are 0.40–0.53 of Fez's, fitted over the same depths with the same decoder. The distance ratios stay below one, $\Lambda_{\mathrm{X}} = 0.69 \pm 0.04$ and

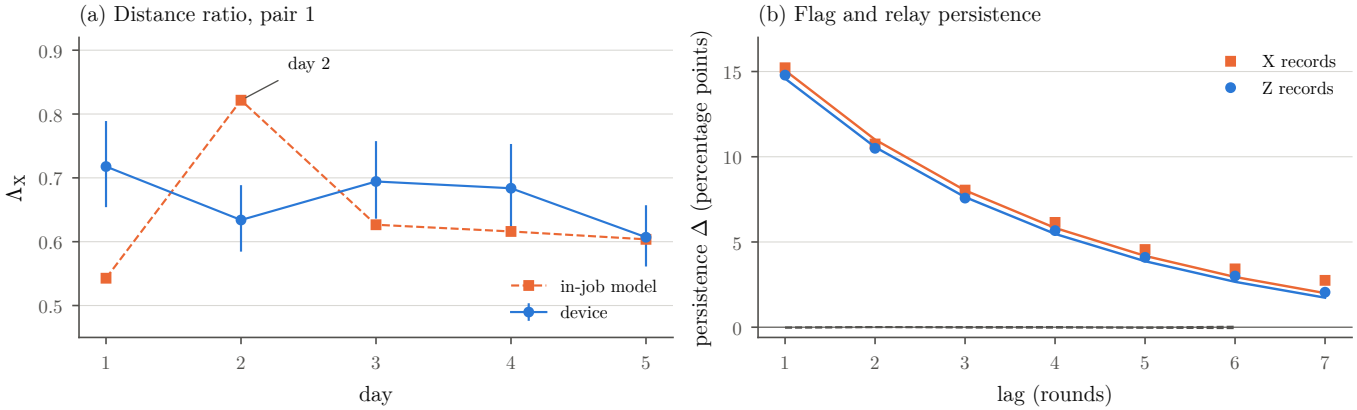


FIG. 5. (a) Pair 1's Λ_X by day, with in-job priors on both sides; device bars are 68% profile intervals. (b) Mean flag and relay persistence with the fitted hidden-Markov curves; the Pauli model's records sit near zero. Shared qubits and overlapping records make neither panel fully independent.

$\Lambda_Z = 0.57 \pm 0.03$ (fit errors only).

Measured idle inputs. For each data qubit the X/Y idle test now fits

$$\ln v(n) = c + \ln G(n) - t_{\mathrm{eq}}(n)/T_q, \tag{15}$$

where t_{eq} is the time the qubit spends on the equator and G holds the modeled gate, readout and Z-axis relaxation losses. The median measured T_q is 0.74 of IBM's T_2, but four qubits reported at 16–24 µs show 59–196 µs under XY4. Eleven of the twenty-five rates moved by more than three standard errors between the two jobs, 35 minutes apart.

With measured T_q and a fixed decoder, the pooled shortfalls are 2.7–10.0 percentage points in X memory and 0.2–2.9 in Z, and a few single-job Z points fall below the model. Similar fractional errors at both distances cancel in the distance ratio,

$$\frac{\Lambda^O}{\Lambda^M} = \frac{\varepsilon_3^O/\varepsilon_3^M}{\varepsilon_5^O/\varepsilon_5^M}, \tag{16}$$

with O the device and M the model. The model can therefore get Λ right while the X-memory decay stays about 69% above it.

TABLE 2. Kingston: fitted decay, device/model. The stored prediction uses its own priors on both sides; the rows below it share one fixed decoder built from the measured decay.

	X memory		Z memory	
	$d = 3$	$d = 5$	$d = 3$	$d = 5$
Stored prediction	1.90	1.69	1.43	1.12
IBM inputs	1.89	1.53	1.38	1.12
Measured T_q	1.69	1.69	1.25	1.22
Joint idle fit	1.64	1.63	1.14	1.15
Separate X/Y and Z fits	1.69	1.69	1.02	1.08

One idle model does not fit both bases. A Z idle test prepares 0 or 1 and reads Z; the median flip probability is 5.3% per round. Each idle test mainly fixes the combination $t_Z/T_1 + t_{\mathrm{eq}}/T_q$, so the two rates come out strongly anticorrelated (median correlation −0.977). A joint fit with non-negative relaxation and pure dephasing puts seventeen of the twenty-five qubits on a constraint boundary; without the constraints, fourteen need negative pure dephasing, seven of them by more than two standard errors. The constrained fit underestimates the Z-idle decay by 11% and overestimates the X-idle decay by 3%. Fitting each basis separately brings Z memory close to the model (Table 2, last row), but with different effective noise in each basis, so there is no single physical picture. The tests were not interleaved, so drift may also play a part.

Readout calibration. The first calibration listed q72 and q12 at 18–20% readout error; the idle offsets and the calibration 88 minutes later put them near 5%. Re-predicting with the later values, with the decoder fixed, leaves gaps of 4.6–11.6 percentage points in X and 3.1–5.5 in Z, larger than with the original inputs. Single calibration entries move the prediction this much.

Raw readout signals. Ten flag, relay and X-ancilla qubits ran thirty cycles of H, a 0.41 µs wait, H, readout and reset, with a 4.87 µs period, 3000 shots and the raw IQ signal recorded. One arm added CZs with each qubit's usual partners and another dropped the resets. The arms ran as separate jobs, the CZ arm about eight minutes after the reference.

Only 0.02–0.18% of the readouts fall outside both reference clouds. In the fitted persistent state most signals stay inside the clouds, and about 60% read 1 (Fig. 6c). The state lives a median 40 µs without CZs and 32 µs with them, close to Fez's 31 µs. Without a calibrated $|2\rangle$ cloud this does not rule out leakage of the measured qubit, and a similar lifetime does not prove the same cause on both chips.

With CZs, qubits that share a partner enter the state

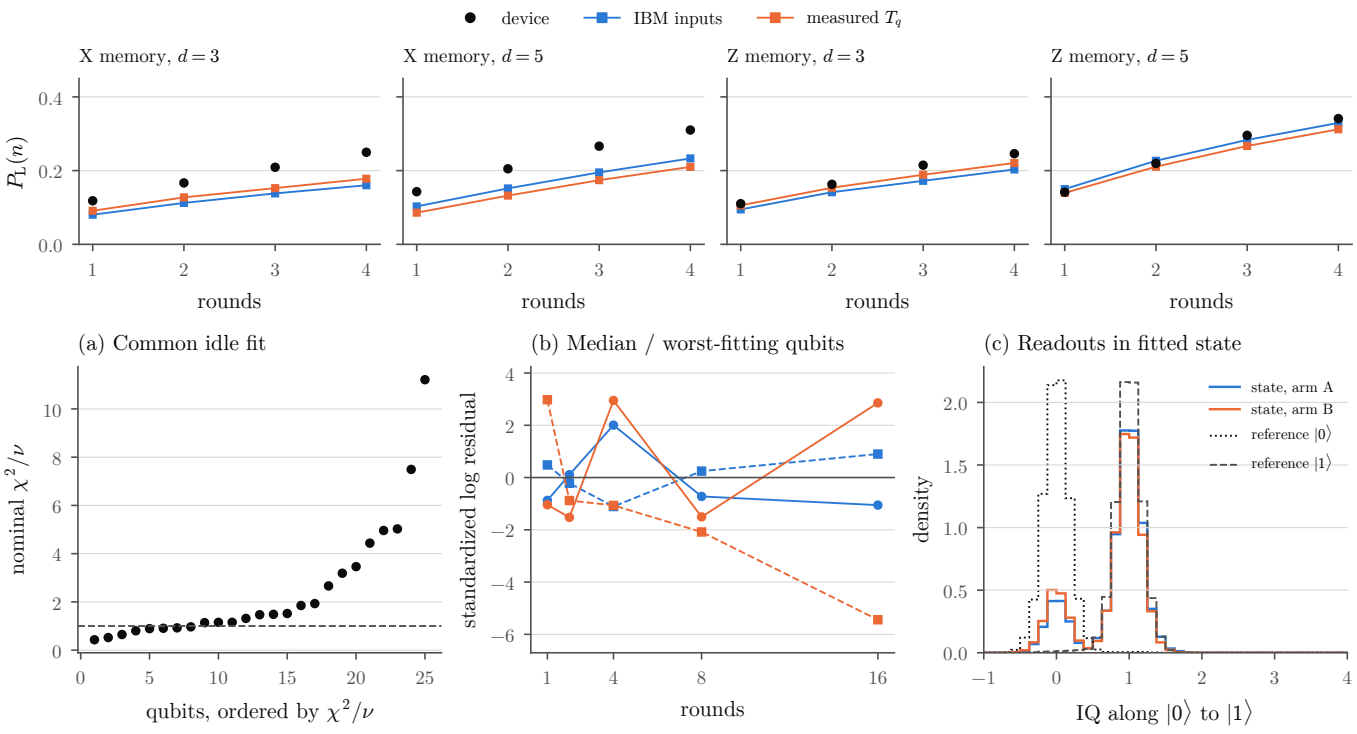


FIG. 6. Kingston. Top: failure probability, decoded for device and models with one capped decoder built from the pooled measured T_q; these updated inputs differ from the stored prediction. Bottom: (a) reduced χ^2 of the joint idle fit for each data qubit; (b) standardized log residuals for a median and the worst qubit; (c) raw readout signal in the fitted persistent state, projected on each qubit's 0-to-1 axis. Most of it stays inside the reference clouds; no $|2\rangle$ reference was available.

together (median correlation 0.13, against 0.009 for other pairs), and the onset rises by about 0.09% per CZ. Without CZs the state is there from the first readout, and its response to the fixed wait fits a frequency shift without revealing its source. In the reset-free memory, flags still persist while relays show no excess.

8. TAKEAWAYS

- The calibration-based Pauli model underestimates the logical error on both chips: by about 1.2–1.3 in X memory and up to 2 in Z memory on Fez, and by about 1.7 in X memory on Kingston even with measured idle decay.
- Distance 5 lost to distance 3 on both chips, as expected for these circuits and placements. Because similar fractional errors cancel, Λ hides most of the model's error.
- On Kingston, dropping the resets and adding XY4 and twirling went with roughly halved decay per round, but the chip and placement changed too.
- Flags and relays carry a persistent state the model lacks. On Kingston it is mostly not a third readout cloud, and CZ partners raise its onset; its source is still open.

CODE AND DATA

The circuits, calibrations, stored predictions, shots and analysis scripts, with reproduction instructions, are at `https://github.com/GoofyGgoober/heavyhex-subsystem-code` (tag `paper`).

[1] C. Chamberland, G. Zhu, T. J. Yoder, J. B. Hertzberg, and A. W. Cross, Topological and subsystem codes on low-degree graphs with flag qubits, Phys. Rev. X **10**, 011022 (2020), arXiv:1907.09528.
[2] D. Bacon, Operator quantum error-correcting subsystems for self-correcting quantum memories, Phys. Rev. A **73**, 012340 (2006), arXiv:quant-ph/0506023.
[3] N. Sundaresan, T. J. Yoder, Y. Kim, M. Li, E. H. Chen, G. Harper, T. Thorbeck, A. W. Cross, A. D. Córcoles, and M. Takita, Demonstrating multi-round subsystem quantum error correction using matching and maximum likelihood decoders, Nat. Commun. **14**, 2852

(2023), arXiv:2203.07205.
[4] J. Napp and J. Preskill, Optimal Bacon–Shor codes, Quantum Inf. Comput. **13**, 490 (2013), arXiv:1209.0794.
[5] C. Gidney, Stim: a fast stabilizer circuit simulator, Quantum **5**, 497 (2021), arXiv:2103.02202.
[6] O. Higgott and C. Gidney, Sparse Blossom: correcting a million errors per core second with minimum-weight matching, Quantum **9**, 1600 (2025), arXiv:2303.15933.
[7] C. Gidney, Stim, version 1.16.0, with the documentation of its `shortest_graphlike_error` and `search_for_undetectable_logical_errors` methods, `https://github.com/quantumlib/Stim` (2026).
[8] R. Harper, C. Lainé, E. T. Hockings, C. K. McLauchlan, G. M. Nixon, B. J. Brown, and S. D. Bartlett, Characterising the failure mechanisms of error-corrected quantum logic gates, Nat. Commun. **17**, 5039 (2026), arXiv:2504.07258.
[9] M. A. Nielsen, A simple formula for the average gate fidelity of a quantum dynamical operation, Phys. Lett. A **303**, 249 (2002), arXiv:quant-ph/0205035.
[10] J. Ghosh, A. G. Fowler, and M. R. Geller, Surface code with decoherence: an analysis of three superconducting architectures, Phys. Rev. A **86**, 062318 (2012), arXiv:1210.5799.
[11] L. S. Shapley, A value for n-person games, in *Contributions to the Theory of Games II*, Annals of Mathematics Studies, Vol. 28, edited by H. W. Kuhn and A. W. Tucker (Princeton University Press, 1953) pp. 307–317.
[12] Google Quantum AI, Exponential suppression of bit or phase errors with cyclic error correction, Nature **595**, 383 (2021), arXiv:2102.06132.
[13] C. C. Bultink *et al.*, Protecting quantum entanglement from leakage and qubit errors via repetitive parity measurements, Sci. Adv. **6**, eaay3050 (2020), arXiv:1905.12731.
[14] B. M. Varbanov, F. Battistel, B. M. Tarasinski, V. P. Ostroukh, T. E. O'Brien, L. DiCarlo, and B. M. Terhal, Leakage detection for a transmon-based surface code, npj Quantum Inf. **6**, 102 (2020), arXiv:2002.07119.